SELF-TEST

Answers are at the end of the chapter.

1. $2 \times 10^1 + 8 \times 10^0$ is equal to
   (a) 10 **(b)** 280 **(c)** 2.8 **(d)** 28
2. The binary number 1101 is equal to the decimal number
   (a) 13 **(b)** 49 **(c)** 11 **(d)** 3
3. The binary number 11011101 is equal to the decimal number
   (a) 121 **(b)** 221 **(c)** 441 **(d)** 256
4. The decimal number 17 is equal to the binary number
   (a) 10010 **(b)** 11000 **(c)** 10001 **(d)** 01001
5. The decimal number 175 is equal to the binary number
   (a) 11001111 **(b)** 10101110 **(c)** 10101111 **(d)** 11101111
6. The sum of 11010 + 01111 equals
   (a) 101001 **(b)** 101010 **(c)** 110101 **(d)** 101000
7. The difference of 110 − 010 equals
   (a) 001 **(b)** 010 **(c)** 101 **(d)** 100
8. The 1's complement of 10111001 is
   (a) 01000111 **(b)** 01000110 **(c)** 11000110 **(d)** 10101010
9. The 2's complement of 11001000 is
   (a) 00110111 **(b)** 00110001 **(c)** 01001000 **(d)** 00111000
10. The decimal number +122 is expressed in the 2's complement form as
    (a) 01111010 **(b)** 11111010 **(c)** 01000101 **(d)** 10000101
11. The decimal number −34 is expressed in the 2's complement form as
    (a) 01011110 **(b)** 10100010 **(c)** 11011110 **(d)** 01011101
12. A single-precision floating-point binary number has a total of
    (a) 8 bits **(b)** 16 bits **(c)** 24 bits **(d)** 32 bits
13. In the 2's complement form, the binary number 10010011 is equal to the decimal number
    (a) −19 **(b)** +109 **(c)** +91 **(d)** −109
14. The binary number 101100111001010100001 can be written in octal as
    (a) 5471230_8 **(b)** 5471241_8 **(c)** 2634521_8 **(d)** 23162501_8
15. The binary number 10001101010001101111 can be written in hexadecimal as
    (a) $AD467_{16}$ **(b)** $8C46F_{16}$ **(c)** $8D46F_{16}$ **(d)** $AE46F_{16}$
16. The binary number for $F7A9_{16}$ is
    (a) 1111011110101001 **(b)** 1110111110101001
    (c) 1111111010110001 **(d)** 1111011010101001
17. The BCD number for decimal 473 is
    (a) 111011010 **(b)** 110001110011 **(c)** 010001110011 **(d)** 010011110011
18. Refer to Table 2–7. The word STOP in ASCII is
    (a) 1010011101010010011111010000 **(b)** 1010010100110010011101010000
    (c) 1001010110110110011101010001 **(d)** 1010011101010010011101100100
19. The number of parity bits to be added to an 8-bit word for constructing Hamming code for detection and correction of single error is
    (a) 1 **(b)** 2 **(c)** 3 **(d)** 4
20. A 7-bit Hamming code (even parity) 0001001 for a BCD digit is known to have single error the encoded BCD digit is
    (a) 9 **(b)** 5 **(c)** 3 **(d)** 0

PROBLEMS

Answers to odd-numbered problems are at the end of the book.

SECTION 2–1 Decimal Numbers

1. What is the weight of the digit 6 in each of the following decimal numbers?
(a) 1386 **(b)** 54,692 **(c)** 671,920

2. Express each of the following decimal numbers as a power of ten:
(a) 10 **(b)** 100 **(c)** 10,000 **(d)** 1,000,000

3. Give the value of each digit in the following decimal numbers:
(a) 471 **(b)** 9356 **(c)** 125,000

4. How high can you count with four decimal digits?

SECTION 2–2 Binary Numbers

5. Convert the following binary numbers to decimal:
(a) 11 **(b)** 100 **(c)** 111 **(d)** 1000
(e) 1001 **(f)** 1100 **(g)** 1011 **(h)** 1111

6. Convert the following binary numbers to decimal:
(a) 1110 **(b)** 1010 **(c)** 11100 **(d)** 10000
(e) 10101 **(f)** 11101 **(g)** 10111 **(h)** 11111

7. Convert each binary number to decimal:
(a) 110011.11 **(b)** 101010.01 **(c)** 1000001.111
(d) 1111000.101 **(e)** 1011100.10101 **(f)** 1110001.0001
(g) 1011010.1010 **(h)** 1111111.11111

8. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?
(a) two **(b)** three **(c)** four **(d)** five **(e)** six
(f) seven **(g)** eight **(h)** nine **(i)** ten **(j)** eleven

9. How many bits are required to represent the following decimal numbers?
(a) 17 **(b)** 35 **(c)** 49 **(d)** 68
(e) 81 **(f)** 114 **(g)** 132 **(h)** 205

10. Generate the binary sequence for each decimal sequence:
(a) 0 through 7 **(b)** 8 through 15 **(c)** 16 through 31
(d) 32 through 63 **(e)** 64 through 75

SECTION 2–3 Decimal-to-Binary Conversion

11. Convert each decimal number to binary by using the sum-of-weights method:
(a) 10 **(b)** 17 **(c)** 24 **(d)** 48
(e) 61 **(f)** 93 **(g)** 125 **(h)** 186

12. Convert each decimal fraction to binary using the sum-of-weights method:
(a) 0.32 **(b)** 0.246 **(c)** 0.0981

13. Convert each decimal number to binary using repeated division by 2:
(a) 15 **(b)** 21 **(c)** 28 **(d)** 34
(e) 40 **(f)** 59 **(g)** 65 **(h)** 73

14. Convert each decimal fraction to binary using repeated multiplication by 2:
(a) 0.98 **(b)** 0.347 **(c)** 0.9028

SECTION 2–4 Binary Arithmetic

15. Add the binary numbers:
    (a) 11 + 01 (b) 10 + 10 (c) 101 + 11
    (d) 111 + 110 (e) 1001 + 101 (f) 1101 + 1011
16. Use direct subtraction on the following binary numbers:
    (a) 11 − 1 (b) 101 − 100 (c) 110 − 101
    (d) 1110 − 11 (e) 1100 − 1001 (f) 11010 − 10111
17. Perform the following binary multiplications:
    (a) 11 × 11 (b) 100 × 10 (c) 111 × 101
    (d) 1001 × 110 (e) 1101 × 1101 (f) 1110 × 1101
18. Divide the binary numbers as indicated:
    (a) 100 ÷ 10 (b) 1001 ÷ 11 (c) 1100 ÷ 100

SECTION 2–5 1's and 2's Complements of Binary Numbers

19. Determine the 1's complement of each binary number:
    (a) 101 (b) 110 (c) 1010
    (d) 11010111 (e) 1110101 (f) 00001
20. Determine the 2's complement of each binary number using either method:
    (a) 10 (b) 111 (c) 1001 (d) 1101
    (e) 11100 (f) 10011 (g) 10110000 (h) 00111101

SECTION 2–6 Signed Numbers

21. Express each decimal number in binary as an 8-bit sign-magnitude number:
    (a) +29 (b) −85 (c) +100 (d) −123
22. Express each decimal number as an 8-bit number in the 1's complement form:
    (a) −34 (b) +57 (c) −99 (d) +115
23. Express each decimal number as an 8-bit number in the 2's complement form:
    (a) +12 (b) −68 (c) +101 (d) −125
24. Determine the decimal value of each signed binary number in the sign-magnitude form:
    (a) 10011001 (b) 01110100 (c) 10111111
25. Determine the decimal value of each signed binary number in the 1's complement form:
    (a) 10011001 (b) 01110100 (c) 10111111
26. Determine the decimal value of each signed binary number in the 2's complement form:
    (a) 10011001 (b) 01110100 (c) 10111111
27. Express each of the following sign-magnitude binary numbers in single-precision floating-point format:
    (a) 0111110000101011 (b) 100110000011000
28. Determine the values of the following single-precision floating-point numbers:
    (a) 1 10000001 01001001110001000000000
    (b) 0 11001100 10000111110100100000000

SECTION 2–7 Arithmetic Operations with Signed Numbers

29. Convert each pair of decimal numbers to binary and add using the 2's complement form:
    (a) 33 and 15 (b) 56 and −27 (c) −46 and 25 (d) −110 and −84
30. Perform each addition in the 2's complement form:
    (a) 00010110 + 00110011 (b) 01110000 + 10101111

31. Perform each addition in the 2's complement form:
(a) 10001100 + 00111001 **(b)** 11011001 + 11100111

32. Perform each subtraction in the 2's complement form:
(a) 00110011 − 00010000 **(b)** 01100101 − 11101000

33. Multiply 01101010 by 11110001 in the 2's complement form.

34. Divide 01000100 by 00011001 in the 2's complement form.

SECTION 2–8 Hexadecimal Numbers

35. Convert each hexadecimal number to binary:
(a) 38_{16} **(b)** 59_{16} **(c)** $A14_{16}$ **(d)** $5C8_{16}$
(e) 4100_{16} **(f)** $FB17_{16}$ **(g)** $8A9D_{16}$

36. Convert each binary number to hexadecimal:
(a) 1110 **(b)** 10 **(c)** 10111
(d) 10100110 **(e)** 1111110000 **(f)** 100110000010

37. Convert each hexadecimal number to decimal:
(a) 23_{16} **(b)** 92_{16} **(c)** $1A_{16}$ **(d)** $8D_{16}$
(e) $F3_{16}$ **(f)** EB_{16} **(g)** $5C2_{16}$ **(h)** 700_{16}

38. Convert each decimal number to hexadecimal:
(a) 8 **(b)** 14 **(c)** 33 **(d)** 52
(e) 284 **(f)** 2890 **(g)** 4019 **(h)** 6500

39. Perform the following additions:
(a) $37_{16} + 29_{16}$ **(b)** $A0_{16} + 6B_{16}$ **(c)** $FF_{16} + BB_{16}$

40. Perform the following subtractions:
(a) $51_{16} - 40_{16}$ **(b)** $C8_{16} - 3A_{16}$ **(c)** $FD_{16} - 88_{16}$

SECTION 2–9 Octal Numbers

41. Convert each octal number to decimal:
(a) 12_8 **(b)** 27_8 **(c)** 56_8 **(d)** 64_8 **(e)** 103_8
(f) 557_8 **(g)** 163_8 **(h)** 1024_8 **(i)** 7765_8

42. Convert each decimal number to octal by repeated division by 8:
(a) 15 **(b)** 27 **(c)** 46 **(d)** 70
(e) 100 **(f)** 142 **(g)** 219 **(h)** 435

43. Convert each octal number to binary:
(a) 13_8 **(b)** 57_8 **(c)** 101_8 **(d)** 321_8 **(e)** 540_8
(f) 4653_8 **(g)** 13271_8 **(h)** 45600_8 **(i)** 100213_8

44. Convert each binary number to octal:
(a) 111 **(b)** 10 **(c)** 110111
(d) 101010 **(e)** 1100 **(f)** 1011110
(g) 101100011001 **(h)** 10110000011 **(i)** 111111101111000

SECTION 2–10 Binary Coded Decimal (BCD)

45. Convert each of the following decimal numbers to 8421 BCD:
(a) 10 **(b)** 13 **(c)** 18 **(d)** 21 **(e)** 25 **(f)** 36
(g) 44 **(h)** 57 **(i)** 69 **(j)** 98 **(k)** 125 **(l)** 156

46. Convert each of the decimal numbers in Problem 45 to straight binary, and compare the number of bits required with that required for BCD.

47. Convert the following decimal numbers to BCD:
    (a) 104 (b) 128 (c) 132 (d) 150 (e) 186
    (f) 210 (g) 359 (h) 547 (i) 1051

48. Convert each of the BCD numbers to decimal:
    (a) 0001 (b) 0110 (c) 1001
    (d) 00011000 (e) 00011001 (f) 00110010
    (g) 01000101 (h) 10011000 (i) 100001110000

49. Convert each of the BCD numbers to decimal:
    (a) 10000000 (b) 001000110111
    (c) 001101000110 (d) 010000100001
    (e) 011101010100 (f) 100000000000
    (g) 100101111000 (h) 0001011010000011
    (i) 1001000000011000 (j) 0110011001100111

50. Add the following BCD numbers:
    (a) 0010 + 0001 (b) 0101 + 0011
    (c) 0111 + 0010 (d) 1000 + 0001
    (e) 00011000 + 00010001 (f) 01100100 + 00110011
    (g) 01000000 + 01000111 (h) 10000101 + 00010011

51. Add the following BCD numbers:
    (a) 1000 + 0110 (b) 0111 + 0101
    (c) 1001 + 1000 (d) 1001 + 0111
    (e) 00100101 + 00100111 (f) 01010001 + 01011000
    (g) 10011000 + 10010111 (h) 010101100001 + 011100001000

52. Convert each pair of decimal numbers to BCD, and add as indicated:
    (a) 4 + 3 (b) 5 + 2 (c) 6 + 4 (d) 17 + 12
    (e) 28 + 23 (f) 65 + 58 (g) 113 + 101 (h) 295 + 157

SECTION 2–11 Digital Codes

53. In a certain application a 4-bit binary sequence cycles from 1111 to 0000 periodically. There are four-bit changes, and because of circuit delays, these changes may not occur at the same instant. For example, if the LSB changes first, the number will appear as 1110 during the transition from 1111 to 0000 and may be misinterpreted by the system. Illustrate how the Gray code avoids this problem.

54. Convert each binary number to Gray code:
    (a) 11011 (b) 1001010 (c) 1111011101110

55. Convert each Gray code to binary:
    (a) 1010 (b) 00010 (c) 11000010001

56. Convert each of the following decimal numbers to ASCII. Refer to Table 2–7.
    (a) 1 (b) 3 (c) 6 (d) 10 (e) 18
    (f) 29 (g) 56 (h) 75 (i) 107

57. Determine each ASCII character. Refer to Table 2–7.
    (a) 0011000 (b) 1001010 (c) 0111101
    (d) 0100011 (e) 0111110 (f) 1000010

58. Decode the following ASCII coded message:

    1001000 1100101 1101100 1101100 1101111 0101110
    0100000 1001000 1101111 1110111 0100000 1100001
    1110010 1100101 0100000 1111001 1101111 1110101
    0111111

59. Write the message in Problem 58 in hexadecimal.

60. Convert the following computer program statement to ASCII:

```
30 INPUT A, B
```

SECTION 2–12 **Error-Detection and Correction Codes**

61. Determine which of the following even parity codes are in error:
(a) 100110010 (b) 011101010 (c) 10111111010001010

62. Determine which of the following odd parity codes are in error:
(a) 11110110 (b) 00110001 (c) 01010101010101010

63. Attach the proper even parity bit to each of the following bytes of data:
(a) 10100100 (b) 00001001 (c) 11111110

64. Attach the proper odd parity bit to each of the bytes of Problem 63.

65. Determine the single-error-correcting code for the following BCD numbers, using odd parity:
(a) 0000 (b) 0001 (c) 0010 (d) 1000

66. 8421 codes are transmitted as Hamming codes with even parity and the following words are received:
(a) 0101000 (b) 0011101 (c) 1100100 (d) 1101001
Find the words that have single error and determine the correct BCD digit.

ANSWERS

SECTION REVIEWS

SECTION 2–1 **Decimal Numbers**

1. (a) 1370: 10 (b) 6725: 100 (c) 7051: 1000 (d) 58.72: 0.1
2. (a) $51 = (5 \times 10) + (1 \times 1)$ (b) $137 = (1 \times 100) + (3 \times 10) + (7 \times 1)$
(c) $1492 = (1 \times 1000) + (4 \times 100) + (9 \times 10) + (2 \times 1)$
(d) $106.58 = (1 \times 100) + (0 \times 10) + (6 \times 1) + (5 \times 0.1) + (8 \times 0.01)$

SECTION 2–2 **Binary Numbers**

1. $2^8 - 1 = 255$
2. Weight is 16.
3. $10111101.011 = 189.375$

SECTION 2–3 **Decimal-to-Binary Conversion**

1. (a) $23 = 10111$ (b) $57 = 111001$ (c) $45.5 = 101101.1$
2. (a) $14 = 1110$ (b) $21 = 10101$ (c) $0.375 = 0.011$

SECTION 2–4 **Binary Arithmetic**

1. (a) $1101 + 1010 = 10111$ (b) $10111 + 01101 = 100100$
2. (a) $1101 - 0100 = 1001$ (b) $1001 - 0111 = 0010$
3. (a) $110 \times 111 = 101010$ (b) $1100 \div 011 = 100$

SECTION 2–5 **1's and 2's Complements of Binary Numbers**

1. (a) 1's comp of 00011010 = 11100101 (b) 1's comp of 11110111 = 00001000
(c) 1's comp of 10001101 = 01110010

2. (a) 2's comp of 00010110 = 11101010 (b) 2's comp of 11111100 = 00000100
 (c) 2's comp of 10010001 = 01101111

SECTION 2–6 Signed Numbers

1. Sign-magnitude: +9 = 00001001
2. 1's comp: −33 = 11011110
3. 2's comp: −46 = 11010010
4. Exponent and mantissa

SECTION 2–7 Arithmetic Operations with Signed Numbers

1. Cases of addition: positive number is larger, negative number is larger, both are positive, both are negative
2. 00100001 + 10111100 = 11011101
3. 01110111 − 00110010 = 01000101
4. Sign of product is positive.
5. 00000101 × 01111111 = 01001111011
6. Sign of quotient is negative.
7. 00110000 ÷ 00001100 = 00000100

SECTION 2–8 Hexadecimal Numbers

1. (a) 10110011 = $B3_{16}$ (b) 110011101000 = $CE8_{16}$
2. (a) 57_{16} = 01010111 (b) $3A5_{16}$ = 001110100101
 (c) $F8OB_{16}$ = 1111100000001011
3. $9B30_{16} = 39{,}728_{10}$
4. $573_{10} = 23D_{16}$
5. (a) $18_{16} + 34_{16} = 4C_{16}$ (b) $3F_{16} + 2A_{16} = 69_{16}$
6. $75_{16} - 21_{16} = 54_{16}$ (b) $94_{16} - 5C_{16} = 38_{16}$

SECTION 2–9 Octal Numbers

1. (a) $73_8 = 59_{10}$ (b) $125_8 = 85_{10}$
2. (a) $98_{10} = 142_8$ (b) $163_{10} = 243_8$
3. (a) 46_8 = 100110 (b) 723_8 = 111010011 (c) 5624_8 = 101110010100
4. (a) 110101111 = 657_8 (b) 1001100010 = 1142_8 (c) 10111111001 = 2771_8

SECTION 2–10 Binary Coded Decimal (BCD)

1. (a) 0010: 2 (b) 1000: 8 (c) 0001: 1 (d) 0100: 4
2. (a) 6_{10} = 0110 (b) 15_{10} = 00010101 (c) 273_{10} = 001001110011
 (d) 849_{10} = 100001001001
3. (a) 10001001 = 89_{10} (b) 001001111000 = 278_{10} (c) 000101010111 = 157_{10}
4. A 4-bit sum is invalid when it is greater than 9_{10}.

SECTION 2–11 Digital Codes

1. (a) 1100_2 = 1010 Gray **(b)** 1010_2 = 1111 Gray **(c)** 11010_2 = 10111 Gray

2. (a) 1000 Gray = 1111_2 **(b)** 1010 Gray = 1100_2 **(c)** 11101 Gray = 10110_2

3. (a) K: 1001011 → $4B_{16}$ **(b)** r: 1110010 → 72_{16}
(c) \$: 0100100 → 24_{16} **(d)** +: 0101011 → $2B_{16}$

SECTION 2–12 Error-Detection and Correction Codes

1. (a) *0*1001011 **(b)** *0*1110010 **(c)** *0*0100100 **(d)** *0*0101011

2. (a) Even **(b)** odd

3. (a) 3 **(b)** 4 **(c)** 4 **(d)** 4

SUPPLEMENTARY PROBLEMS FOR EXAMPLES

2–1 9 has a value of 900, 3 has a value of 30, 9 has a value of 9.

2–2 6 has a value of 60, 7 has a value of 7, 9 has a value of 9/10 (0.9), 2 has a value of 2/100 (0.02), 4 has a value of 4/1000 (0.004).

2–3 10010001 = 128 + 16 + 1 = 145 **2–4** 10.111 = 2 + 0.5 + 0.25 + 0.125 = 2.875

2–5 125 = 64 + 32 + 16 + 8 + 4 + 1 = 1111101 **2–6** 39 = 100111

2–7 1111 + 1100 = 11011 **2–8** 111 − 100 = 011 **2–9** 110 − 101 = 001

2–10 1101 × 1010 = 10000010 **2–11** 1100 ÷ 100 = 11 **2–12** 00110101

2–13 01000000 **2–14** See Table 2–17. **2–15** 01110111 = $+119_{10}$

▶ **TABLE 2–17**

	SIGN-MAGNITUDE	1'S COMP	2'S COMP
+19	00010011	00010011	00010011
−19	10010011	11101100	11101101

2–16 11101011 = -20_{10} **2–17** 11010111 = -41_{10}

2–18 1100001000101001100000000 **2–19** 01010101 **2–20** 00010001

2–21 1001000110 **2–22** (83)(−59) = −4897 (10110011011111 in 2's comp)

2–23 100 ÷ 25 = 4 (0100) **2–24** $4F79C_{16}$ **2–25** 0110101111010011_2

2–26 $6BD_{16} = 011010111101 = 2^{10} + 2^9 + 2^7 + 2^5 + 2^4 + 2^3 + 2^2 + 2^0$
$= 1024 + 512 + 128 + 32 + 16 + 8 + 4 + 1 = 1725_{10}$

2–27 $60A_{16} = (6 \times 256) + (0 \times 16) + (10 \times 1) = 1546_{10}$

2–28 $2591_{10} = A1F_{16}$ **2–29** $4C_{16} + 3A_{16} = 86_{16}$ **2–30** $BCD_{16} - 173_{16} = A5A_{16}$

2–31 (a) $001011_2 = 11_{10} = 13_8$ **(b)** $010101_2 = 21_{10} = 25_8$
(c) $001100000_2 = 96_{10} = 140_8$ **(d)** $111101010110_2 = 3926_{10} = 7526_8$

2–32 1250762_8 **2–33** 1001011001110011 **2–34** $82{,}276_{10}$

2–35 1001100101101000 **2–36** 10000010 **2–37 (a)** 111011 (Gray) **(b)** 111010_2

2–38 The sequence of codes for `80 INPUT Y` is $38_{16}30_{16}20_{16}49_{16}4E_{16}50_{16}55_{16}54_{16}20_{16}59_{16}$

2–39 01001011 **2–40** Yes **2–41** 1100110 **2–42** 011110110 **2–43** 3

2–44 111001001

SELF-TEST

1. (d) **2.** (a) **3.** (b) **4.** (c) **5.** (c) **6.** (a) **7.** (d) **8.** (b)
9. (d) **10.** (a) **11.** (c) **12.** (d) **13.** (d) **14.** (b) **15.** (c) **16.** (a)
17. (c) **18.** (a) **19.** (d) **20.** (a)